

IoT: Opt for the Right Open Source Database

By harnessing the right open source database solutions for IoT, organisations can drive innovation, enhance data management, and get actionable insights that propel them ahead in the IoT landscape. Unlock the full potential of IoT by selecting the right set of open source databases.

The technology ecosystem of Internet of Things (IoT) is complex, and includes hardware, connectivity, platforms, middleware, analytics, operating systems, applications, and services.

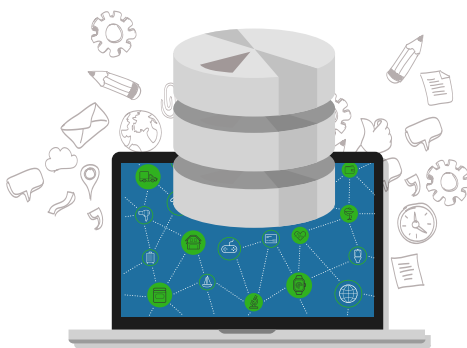
As per IDC, there will be 41.6 billion connected IoT devices or 'things', generating 79.4 zettabytes (ZB) of data in 2025. Sixty-four per cent of those surveyed think open source is significant in deploying IoT, as per the IDC Global IoT Decision-Maker Survey. According to a survey by W3.Org, 91% of IoT developers use open source software, open hardware, or open data in at least one part of their development stack.

According to Statista, the total installed base of Internet of Things (IoT) connected devices is projected to amount to 75.44 billion worldwide by 2025. By 2030, IoT could enable US\$ 5.5 trillion to \$12.6 trillion in value globally, including the value captured by consumers and customers of IoT products and services, as per McKinsey.

With billions of devices generating trillions of bytes of data, it is very crucial for enterprises to organise, store and work with all the data that is generated.

In general, IoT applications leverage both relational and non-relational (NoSQL) databases. The selection of the type of the database depends on the application type. In most of the cases, a combination of both the databases can be utilised.

Most IoT applications are heterogeneous and domain-centric.



Choosing the most efficient database for these applications can be challenging. The important parameters for choosing the right database for IoT applications are scalability, availability, ability to handle huge amounts of data, high processing speed, schema flexibility, integration with varied analytical tools, security and costs.

Open source adoption for IoT

The following are the major drivers that prompt enterprises to adopt open source technologies for IoT.

- **Cost:** Adoption of open source IoT frameworks involves no money, as they are free for use.
- **Efficiency:** Adoption of open source helps developers reduce development times, thereby reducing the development cost.
- **Scalability:** Where billions of devices are supported using open source technology, it helps in affordable scaling.
- **Innovation:** Open source adoption in building newer applications is permissionless and low risk, which

leads to better innovation and agility.

- **Open source API:** Open source APIs are a standard gateway for IoT framework-based applications. They help to improve the communication between various software components, hardware devices, and systems.
- **Libraries:** An open source IoT framework offers a wide range of libraries, SDKs, and open source hardware. Raspberry Pi and Arduino are open source tools for customising IoT platforms.
- **Security:** Open source software can protect individuals' data by implementing powerful encryption (SSH, SSL, PGP, etc).
- **Interoperability:** The adoption of open source helps in collaboration and interoperability.

Drivers for adopting open source databases for IoT

Most IoT solutions are distributed across various geographical locations. This necessitates solutions that adopt fog computing at the edge and cloud computing at the enterprise level. However, no single database product in the market can fulfil the IoT database requirements across the organisation. There is need for a collection of databases, potentially from a variety of vendors, used in one or more stages of the IoT life cycle.

The key business drivers for open source database adoption are:

- Flexibility to process the data at the edge